GENERATIVE AI & CONVERSATIONAL AI

ThangarajAI Chatbot

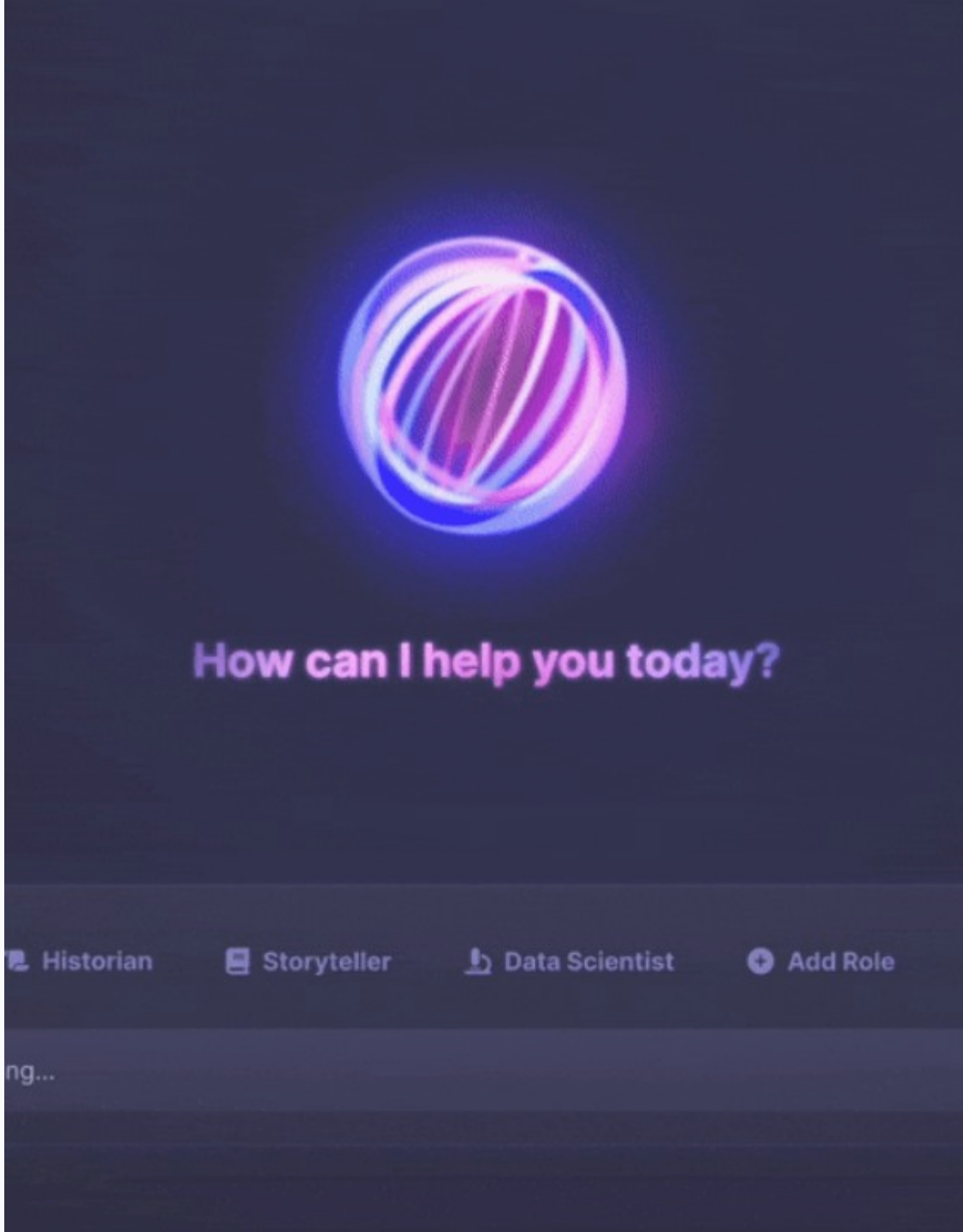
AI-powered learning platform with hybrid FAQ retrieval and Gemini fallback.

github.com/tmthangaraj1980-AIML/AI-PROJECTS/ThangarajAI_Chatbot

THE PROBLEM

Interview Prep Friction

- **Scattered Knowledge:** AI/ML concepts are fragmented across documentation.
- **Slow Retrieval:** Searching manuals wastes valuable study time.
- **Static Repositories:** Pre-written FAQs fail on unmapped edge queries.



Core Capabilities

Curated Knowledge

500+ AI/ML interview questions natively mapped.

Intelligent Matching

Similarity thresholds matching user intent to FAQs.

Gemini Fallback

Real-time generative generation for unknown queries.

Technology Stack



Python



Streamlit



Pandas



Difflib

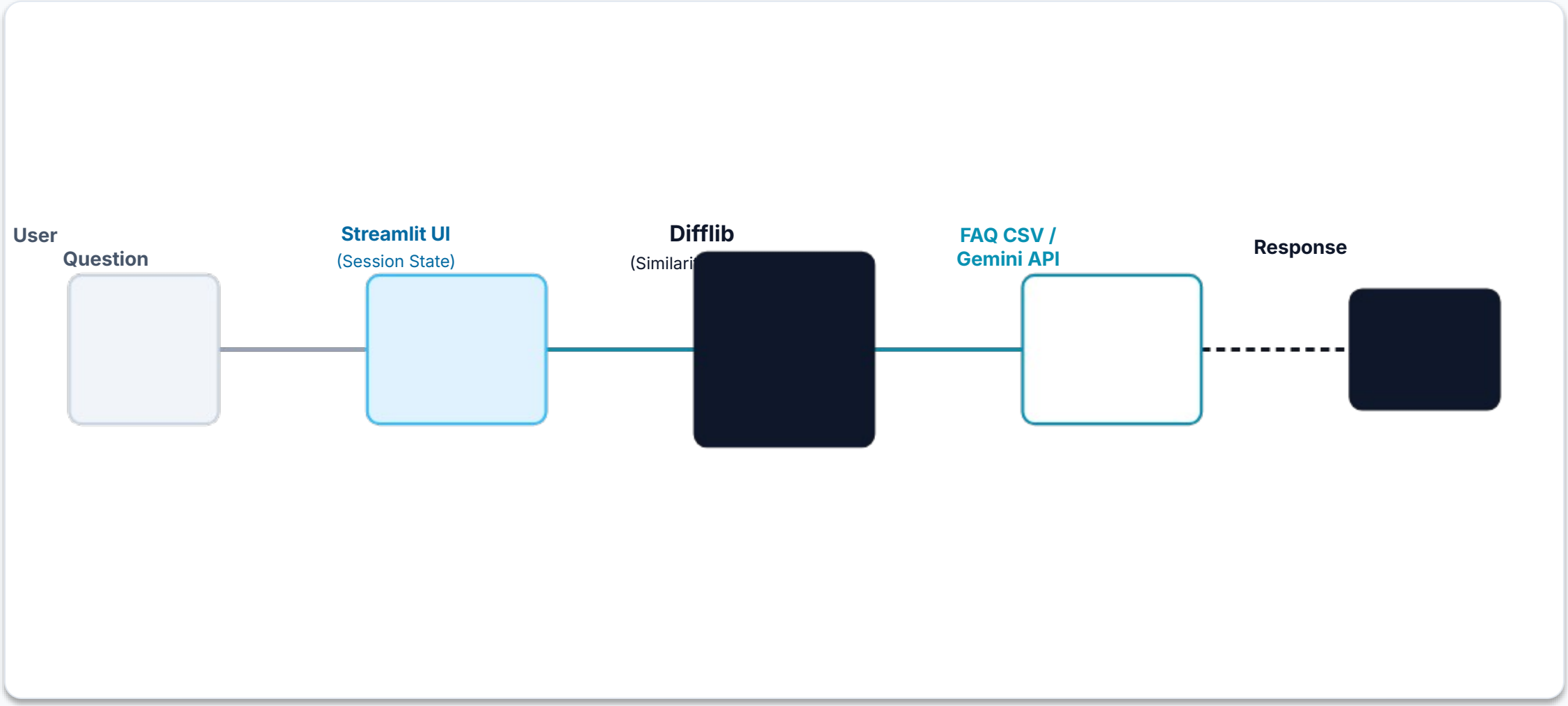


Gemini 2.5 Flash



Transformers

Application Architecture



Inference Workflow

1. Clean
Strip bullets and lowercase query.

2. Match
Evaluate string similarity cutoff.

3. Route
Query Dataset or call Gemini.

4. Render
Streamlit chat output generation.

Domains & Data Pipeline

Knowledge Domains

- Python & Data Science.
- Machine Learning & NLP.
- Deep Learning & Generative AI.

Dataset Mechanics

- 500+ curated FAQ rows loaded via Pandas.
- Lowercase structural normalization mapping.
- O(1) exact match lookup prioritization.

Hybrid Retrieval Engine

String Similarity Lookup

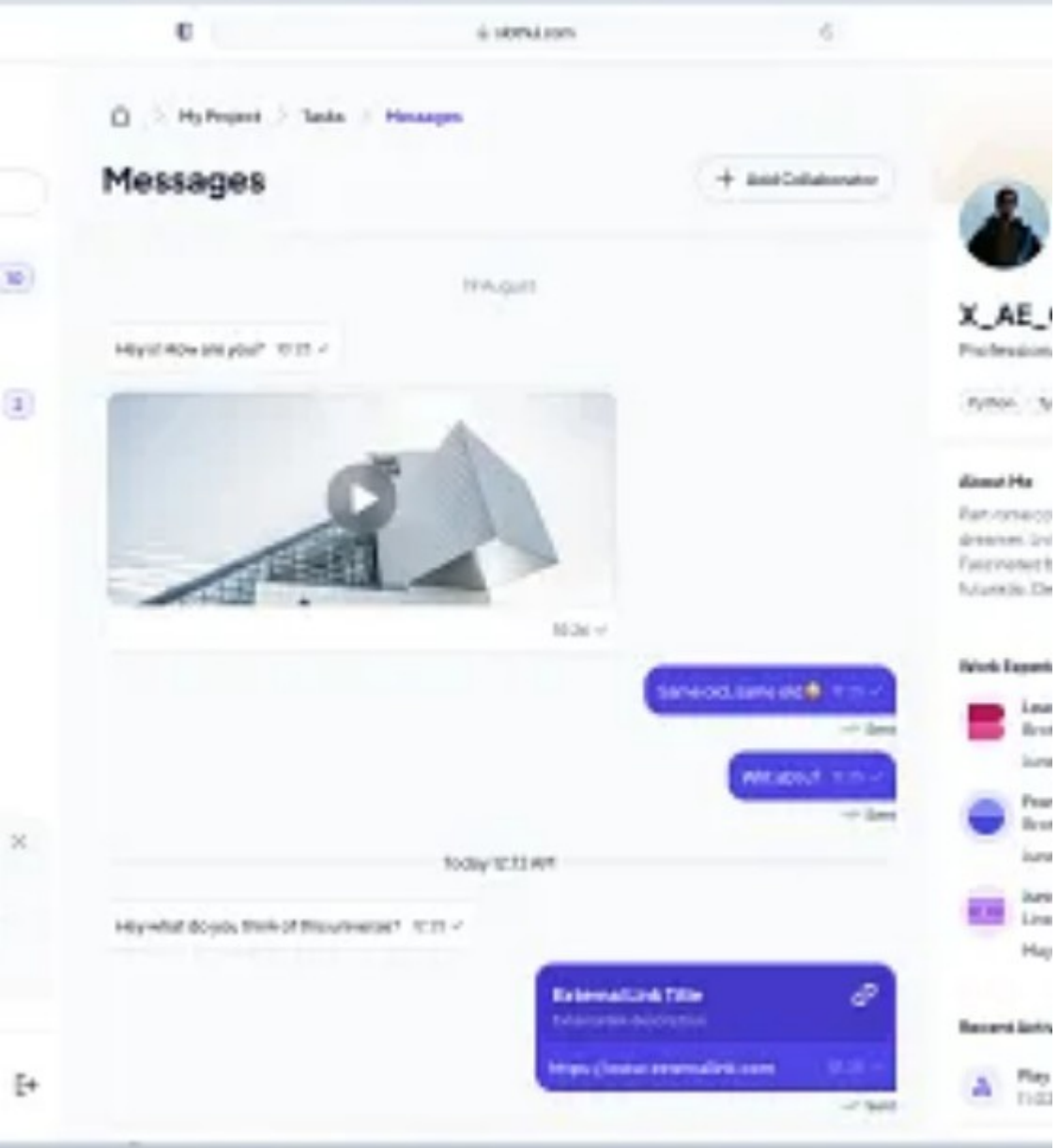
Difflib SequenceMatcher

Evaluates string similarity with a strict 0.75+ cutoff threshold to guarantee high-accuracy local dataset retrieval.

Generative Fallback

Gemini 2.5 Flash API

Handles out-of-scope questions by generating dynamic responses securely via environment-injected API keys.



User Interface

Interactive learning dashboard built completely in Streamlit using conversational UI elements.

- **Streamlit chat_input element.**
- **Dataset metrics header display.**
- **Topic outline sidebar.**

Future Roadmap

Enhanced Interactions

+ Voice Input: Expanding accessibility for learners.
+ AI Mock Interviews: Simulating real-time pressure tests.

Advanced Integrations

+ RAG Architecture: Integration of full PDF knowledge bases.
+ Resume Parsing: Tailoring interview questions to specific CVs.

Technical Q&A

1. Why use Difflib over Vector Search?

Lightweight, zero-latency string matching ideal for exact interview FAQ lookups.

2. How is the fallback triggered?

If the get_close_matches similarity score falls below 0.75, the query routes to Gemini.

3. State management in Streamlit?

Session_state explicitly preserves the current_question and current_response across UI renders.

4. How are user inputs normalized?

Bullet points and hyphens are stripped, and text is lowercased prior to dictionary lookup.

5. Security approach for Gemini API?

Loaded securely via os.getenv() to prevent hardcoding keys in public GitHub repositories.

ThangarajAI - Knowledge Hub

Bridging the gap between static dataset retrieval and generative intelligence.



/AI-PROJECTS/ThangarajAI_Chatbot



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Image Sources



https://files.smashing.media/articles/designing-ai-beyond-conversational-interfaces/9-abstraction-rolebot.gif

Source: www.smashingmagazine.com



https://cdn.dribbble.com/userupload/15008698/file/original-59cd89b153b55320f0c8719207ce5c31.jpg?format=webp&resize=400x300&vertical=center

Source: dribbble.com